

Course: Cloud and Network Security CNS-C2-2026

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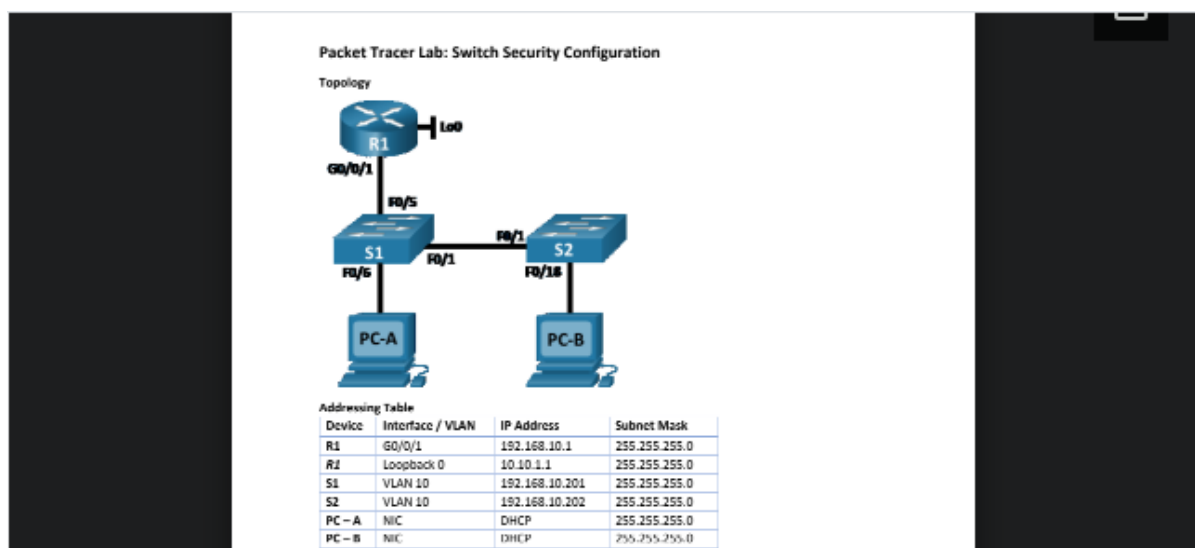
September 30, 2026

Week 4 Assignment 1:

VLANs and Secure Switch Configuration

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Objectives

- Configure the network devices and verify basic settings.
- Configure VLANs on the switches.
- Configure switch security features to protect the Layer 2 network.

Steps

Part 1: Configure the Network Devices

Part 2: Configure VLANs on the Switches

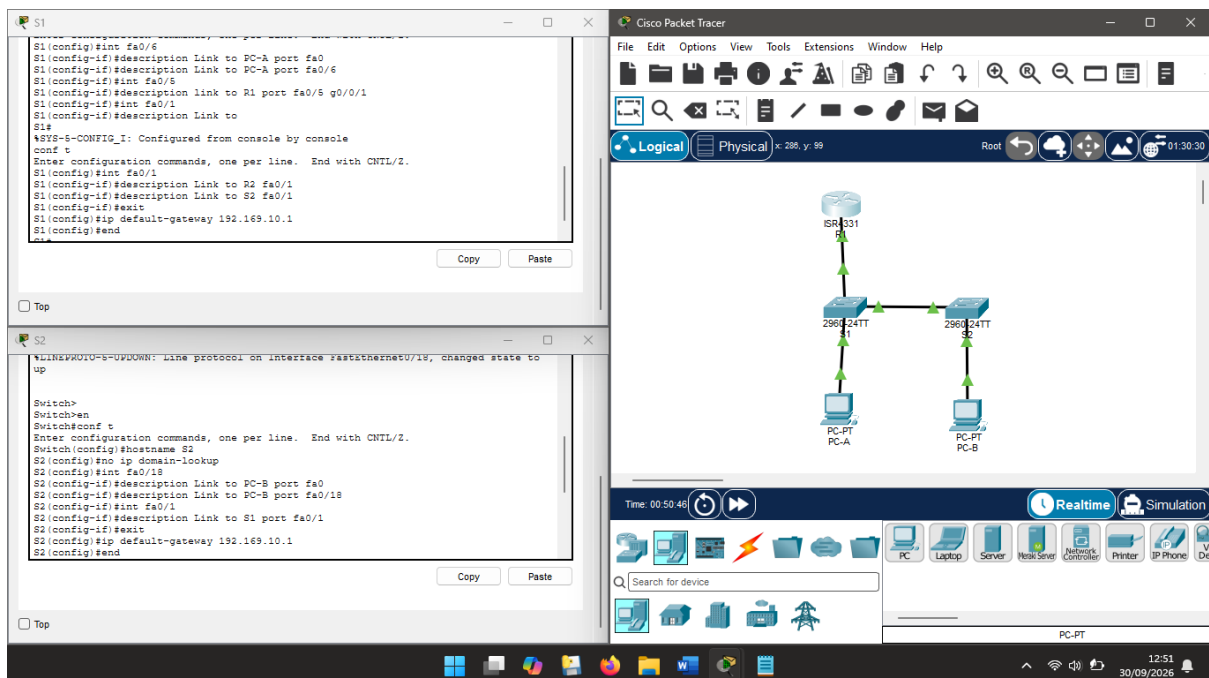
Part 3: Configure Switch Security

Introduction

This report documents the configuration of a small switched network made up of one Cisco router, two Catalyst 2960 switches, and two PCs. The network is segmented with VLANs, and Layer 2 security controls are applied to protect it from common switch-level threats such as unauthorized access, rogue DHCP servers, and spanning-tree manipulation. The work covers basic device setup, VLAN and trunk configuration, and a set of switch hardening measures. It ends with an end-to-end connectivity test to confirm that the design works as intended.

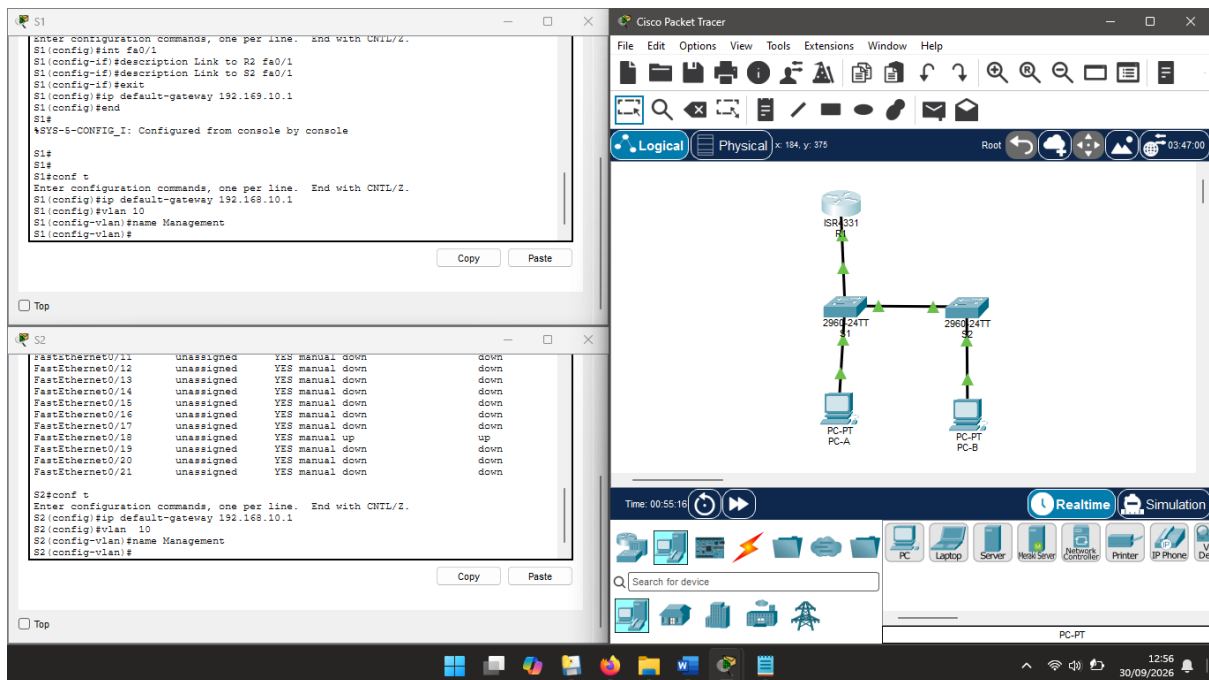
Step 1: Cable the network according to the topology and configure R1.

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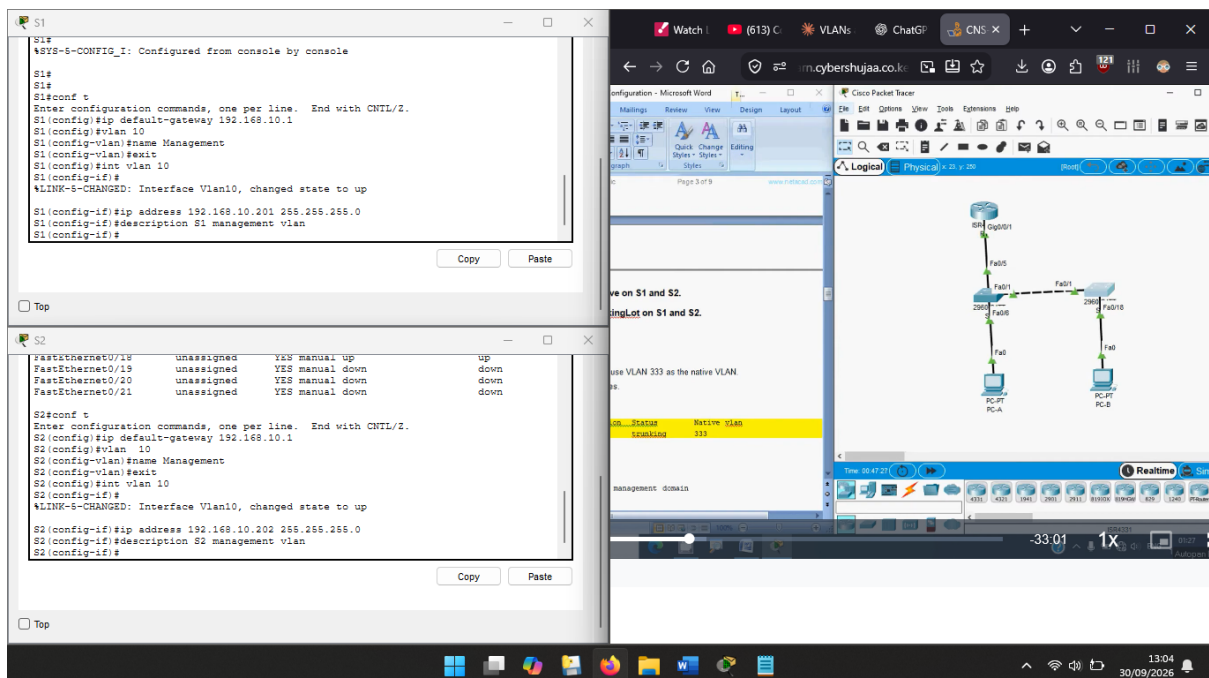


Part 2: Configure VLANs on the Switches

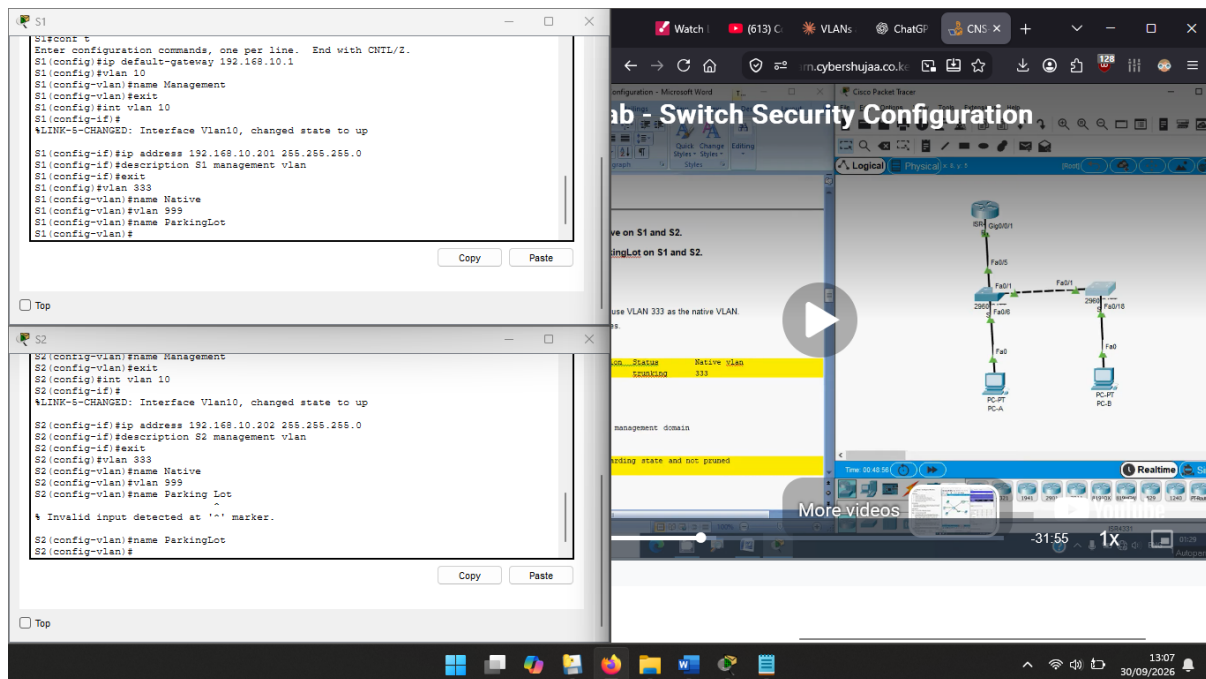
Step 1: Configure VLAN 10.



Step 2: Configure the SVI for VLAN 10.



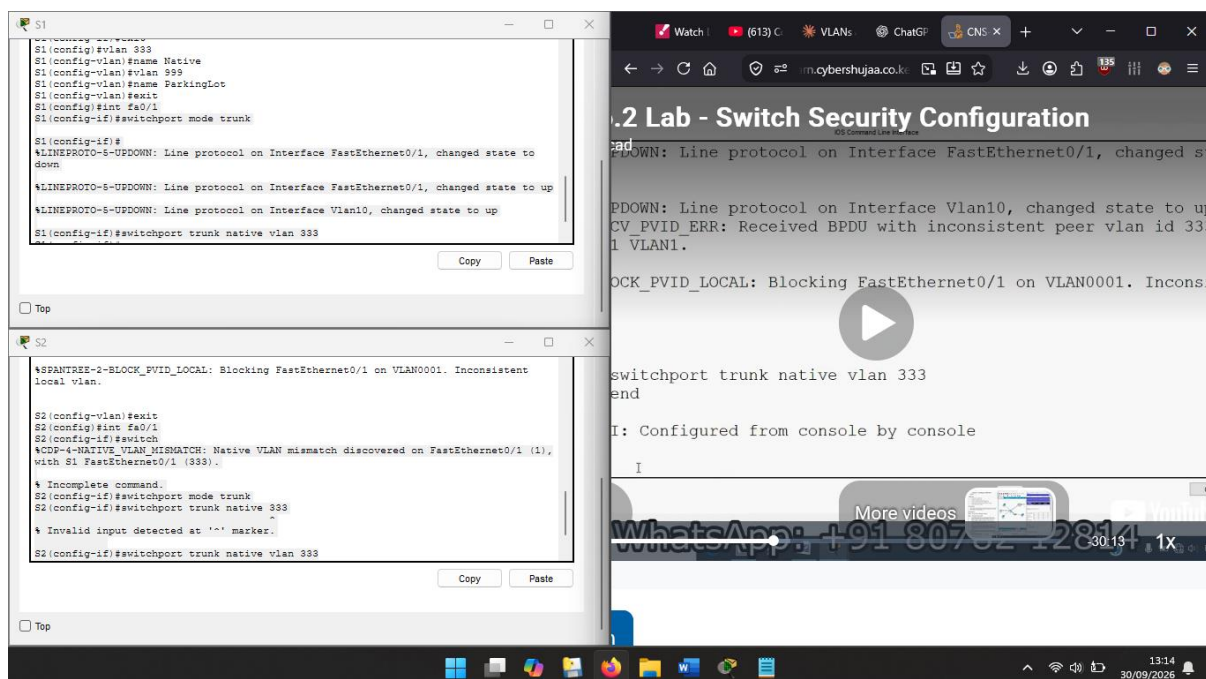
Step 3: Configure VLAN 999 with the name ParkingLot on S1 and S2.



Part 3: Configure Switch Security

Step 1: Implement 802.1Q trunking.

configure trunking on F0/1 to use VLAN 333 as the native VLAN.



show interfaces trunk

The image shows a Cisco Packet Tracer simulation environment. On the left, two configuration windows for switches S1 and S2 are open. S1's configuration shows Fa0/1 as a trunk port with VLANs 1-1005 allowed. S2's configuration shows Fa0/1 as a trunk port with VLANs 1-1005 allowed. The network diagram on the right shows S1 and S2 connected to a central router (ISR 331) and two PCs (PC-A and PC-B).

```
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
S1#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      333
Port      Vlans allowed on trunk
Fa0/1     1-1005
Port      Vlans allowed and active in management domain
Fa0/1     1,10,333,999
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,333,999
S1#
```

```
S2
consistency restored.
S2
S2(config-if)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console
S2#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      333
Port      Vlans allowed on trunk
Fa0/1     1-1005
Port      Vlans allowed and active in management domain
Fa0/1     1,10,333,999
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,333,999
S2#
```

Disable DTP negotiations

The image shows a Cisco Packet Tracer simulation environment. On the left, two configuration windows for switches S1 and S2 are open. S1's configuration shows Fa0/1 as a trunk port with VLANs 1-1005 allowed. S2's configuration shows Fa0/1 as a trunk port with VLANs 1-1005 allowed. The network diagram on the right shows S1 and S2 connected to a central router (ISR 331) and two PCs (PC-A and PC-B).

```
S1
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      333
Port      Vlans allowed on trunk
Fa0/1     1-1005
Port      Vlans allowed and active in management domain
Fa0/1     1,10,333,999
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,333,999
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int fa0/1
S1(config-if)#switchport nonegotiate
S1(config-if)#end
S1#
```

```
S2
Fa0/1     on        802.1q         trunking      333
Port      Vlans allowed on trunk
Fa0/1     1-1005
Port      Vlans allowed and active in management domain
Fa0/1     1,10,333,999
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,333,999
S2#
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#int fa0/1
S2(config-if)#switchport nonegotiate
S2(config-if)#end
S2#
```


Verify with the show interfaces command.

The screenshot shows a video player displaying a Cisco Packet Tracer configuration. The video player interface includes a title bar with 'Watch (613)', 'VLAN', 'ChatG', and 'CN'. The video content shows a configuration for S1 and S2 switches. The S1 configuration is as follows:

```
show interface fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 333 (Native)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
```

The S2 configuration is as follows:

```
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 333 (Native)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
--More--
```

The video player interface also shows a 'More videos' button and a 'Mission status' button.

Step 2: Configure access ports.

The screenshot shows a video player displaying a Cisco Packet Tracer configuration. The video player interface includes a title bar with 'Watch (613)', 'VLAN', 'ChatG', and 'CN'. The video content shows a configuration for S1 and S2 switches. The S1 configuration is as follows:

```
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
S1#
S1#
S1#
S1#
S1#
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int range fa0/5-6
S1(config-if-range)#switchport mode access
S1(config-if-range)#
```

The S2 configuration is as follows:

```
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
S2#
S2#
S2#
S2#
S2#
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#int range fa0/18
S2(config-if-range)#switchport mode access
S2(config-if-range)#
```

The video player interface also shows a 'More videos' button and a 'Mission status' button.

Step 3: Secure and disable unused switchports.

The video player shows a Cisco Packet Tracer lab configuration for Step 3: Secure and disable unused switchports. The video shows two switches, S1 and S2, with their configurations displayed in the terminal windows.

S1 Configuration:

```
S1(config)#int range fa0/2-4,fa0/7-24,g0/1-2
^
Invalid input detected at '^' marker.
S1(config)#int range fa0/2-4,fa0/7-24,g0/1-2
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 999
S1(config-if-range)#shutdown
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
```

S2 Configuration:

```
S2(config)#int range fa0/2-17,fa0/19-24,g0/1-2
S2(config-if-range)#switchport mode access
S2(config-if-range)#switchport access vlan 999
S2(config-if-range)#shutdown
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
```

show interface status

The video player shows a Cisco Packet Tracer lab configuration for Step 3: Secure and disable unused switchports. The video shows two switches, S1 and S2, with their configurations displayed in the terminal windows.

S1 Configuration:

```
S1#show interface status
Port Name Status Vlan Duplex Speed Type
Fa0/1 Link to S2 fa0/1 connected trunk a-full a-100 10/100BaseTX
Fa0/2 disabled 999 auto auto 10/100BaseTX
Fa0/3 disabled 999 auto auto 10/100BaseTX
Fa0/4 link to S1 port fa connected 1 a-full a-100 10/100BaseTX
Fa0/6 Link to PC-A port connected 1 a-full a-100 10/100BaseTX
Fa0/7 disabled 999 auto auto 10/100BaseTX
Fa0/8 disabled 999 auto auto 10/100BaseTX
Fa0/9 disabled 999 auto auto 10/100BaseTX
Fa0/10 disabled 999 auto auto 10/100BaseTX
Fa0/11 disabled 999 auto auto 10/100BaseTX
Fa0/12 disabled 999 auto auto 10/100BaseTX
Fa0/13 disabled 999 auto auto 10/100BaseTX
Fa0/14 disabled 999 auto auto 10/100BaseTX
Fa0/15 disabled 999 auto auto 10/100BaseTX
```

S2 Configuration:

```
S2#show interface status
Port Name Status Vlan Duplex Speed Type
Fa0/1 Link to S1 port fa connected trunk a-full a-100 10/100BaseTX
Fa0/2 disabled 999 auto auto 10/100BaseTX
Fa0/3 disabled 999 auto auto 10/100BaseTX
Fa0/4 disabled 999 auto auto 10/100BaseTX
Fa0/5 disabled 999 auto auto 10/100BaseTX
Fa0/6 disabled 999 auto auto 10/100BaseTX
```

Step 4: Document and implement port security features.

S1 show port-security int fa0/6

```
S1#
S1#
S1#
S1#
S1#show port-security int fa0/6
Port Security          : Disabled
Port Status            : Secure-down
Violation Mode         : Shutdown
Aging Time             : 0 min
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
S1#
S1#
S1#
```

Copy Paste

Top

S1 implement port security features

```
S1#
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int fa0/6
S1(config-if)#switchport port-security
S1(config-if)#switchport port-security max 3
S1(config-if)#switchport port-security vi?
violation
S1(config-if)#switchport port-security vi
S1(config-if)#switchport port-security violation ?
protect Security violation protect mode
restrict Security violation restrict mode
shutdown Security violation shutdown mode
S1(config-if)#switchport port-security violation re?
restrict
S1(config-if)#switchport port-security violation re restrict
S1(config-if)#switchport port-security aging time 60
S1(config-if)#
```

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Top

S1 show port-security int fa0/6 after implement port security

```
S1(config-if)#end
S1#
$SYS-S-CONFIG_I: Configured from console by console
S1#
S1#show port-security int fa0/6
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Restrict
Aging Time             : 60 min
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 3
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
S1#
```

Copy Paste

Top

S2 Configure port security

```
***
S2(config-if)#switch port-security mac-address s
S2(config-if)#switch port-security mac-address sticky
S2(config-if)#
S2(config-if)#
S2(config-if)#switchport port-security max 2
S2(config-if)#switchport port-security violation protect
S2(config-if)#switchport port-security aging 60
^ Invalid input detected at '^' marker.
S2(config-if)#switchport port-security aging 60
^ Unrecognized command
S2(config-if)#switchport port-security aging?
aging
S2(config-if)#switchport port-security aging
S2(config-if)#switchport port-security aging ?
time Port-security aging time
S2(config-if)#switchport port-security aging time 60
S2(config-if)#
S2(config-if)#
```

Copy Paste

Top

S2 show port-security int fa0/6 after implement port security

```
S2(config-if)#switchport port-security aging time 60
S2(config-if)#
S2(config-if)#end
S2#
*SYS-S-CONFIG_I: Configured from console by console
S2#show port-security int fa0/18
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Protect
Aging Time             : 60 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 2
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
S2#
```

Copy Paste

Top

Step 5: Implement DHCP snooping security.

S2 config

```
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#ip dhcp snooping
S2(config)#ip dhcp snooping vlan 10
S2(config)#int fa0/1
S2(config-if)#ip dhcp snooping trust
S2(config-if)#exit
S2(config)#int fa0/18
S2(config-if)#ip dhcp snooping limit rate 5
S2(config-if)#end
S2#
*SYS-S-CONFIG_I: Configured from console by console
show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
-----
Interface      Trusted      Rate limit (pps)
-----
FastEthernet0/1  yes         unlimited
FastEthernet0/18 no           5
S2#
```

Copy Paste

Top

PC-B, release and then renew the IP address.

PC-B

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /release
Port is not using DHCP.
C:\>ipconfig /renew

IP Address. . . . .: 192.168.10.10
Subnet Mask. . . . .: 255.255.255.0
Default Gateway. . . . .: 192.168.10.1
DNS Server. . . . .: 0.0.0.0

C:\>
```

Top

Change all ports to vlan 10

S1

```
show vlan brief
```

VLAN Name	Status	Ports
1 default	active	
10 Management	active	Fa0/5, Fa0/6
333 Native	active	Fa0/2, Fa0/3, Fa0/4, Fa0/7
999 ParkingLot	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11
		Fa0/12, Fa0/13, Fa0/14, Fa0/15
		Fa0/16, Fa0/17, Fa0/18, Fa0/19
		Fa0/20, Fa0/21, Fa0/22, Fa0/23
		Fa0/24, Gig0/1, Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

S2

```
S2#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	
10 Management	active	Fa0/18
333 Native	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
999 ParkingLot	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9
		Fa0/10, Fa0/11, Fa0/12, Fa0/13
		Fa0/14, Fa0/15, Fa0/16, Fa0/17
		Fa0/19, Fa0/20, Fa0/21, Fa0/22
		Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

PC-B

```
Command Prompt
```

```
IP Address.....: 192.168.10.10
Subnet Mask.....: 255.255.255.0
Default Gateway...: 192.168.10.1
DNS Server.....: 0.0.0.0

C:\>ipconfig /release

IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway...: 0.0.0.0
DNS Server.....: 0.0.0.0

C:\>ipconfig /renew
DHCP request failed.

C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>ipconfig /release

IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway...: 0.0.0.0
DNS Server.....: 0.0.0.0

C:\>ipconfig /renew
IP Address.....: 192.168.10.10
Subnet Mask.....: 255.255.255.0
Default Gateway...: 192.168.10.1
DNS Server.....: 0.0.0.0

C:\>
```

Verify the DHCP snooping binding using

S2

```
S2#show ip dhcp snooping binding
```

MacAddress	IpAddress	Lease(sec)	Type	VLAN	Interface
00:E0:F7:5B:E9:E3	192.168.10.10	0	dhcp-snooping	10	FastEthernet0/18

Total number of bindings: 1

S2#

Step 6: Implement PortFast and BPDU guard.

Configure PortFast on all the access ports that are in use on both switches.

The screenshot displays two terminal windows for switches S1 and S2, and a document titled 'Conclusion section'.

S1 Configuration:

```
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int fa0/5-6
^
Invalid input detected at '^' marker.
S1(config)#int range fa0/5-6
S1(config-if-range)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/5 but will only
have effect when the interface is in a non-trunking mode.
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
```

S2 Configuration:

```
S2#
S2#
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#int fa0/18
S2(config-if)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/18 but will only
have effect when the interface is in a non-trunking mode.
S2(config-if)#
```

Conclusion section:

support your answers where necessary.

3. **Conclusion section**, written in your own words, that captures a reflective summary of what you have learned and your experience.

Step 6: Implement PortFast and BPDU guard.

1. Configure PortFast on all the access ports that are in use on both switches.
S1(config)# interface range fa0/5 - 6
S1(config-if)# spanning-tree portfast
S2(config)# interface fa0/18
S2(config-if)# spanning-tree portfast
2. Enable BPDU guard on S1 and S2 VLAN 10 access ports connected to PC-A and PC-B.
S1(config)# interface fa0/6
S1(config-if)# spanning-tree bpduguard enable
S2(config)# interface fa0/18
S2(config-if)# spanning-tree bpduguard enable
3. Verify that BPDU guard and PortFast are enabled on the appropriate ports.

S1# show spanning-tree interface fa0/6 detail
Port 8 (FastEthernet0/6) of VLAN0010 is designated forwarding
Port path cost: 19,210,000, Port priority: 128, Port role: designated forwarding
Output only: 1,000,000,000, Port state: 1
Number of times the port has been in the state: 1
The port is in the spanning-tree mode.
Link type is point-to-point by default

Enable BPDU guard on S1 and S2 VLAN 10 access ports connected to PC-A and PC-B.

The screenshot displays two terminal windows for switches S1 and S2, and a document titled 'Conclusion section'.

S1 Configuration:

```
S1#
S1#
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int fa0/6
S1(config-if)#spanning-tree bpduguard enable
S1(config-if)#
S1(config-if)#
```

S2 Configuration:

```
S2#
S2#
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#int fa0/18
S2(config-if)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/18 but will only
have effect when the interface is in a non-trunking mode.
S2(config-if)#
S2(config-if)#
S2(config-if)#spanning-tree bpduguard enable
S2(config-if)#
```

Conclusion section:

support your answers where necessary.

3. **Conclusion section**, written in your own words, that captures a reflective summary of what you have learned and your experience.

Step 6: Implement PortFast and BPDU guard.

1. Configure PortFast on all the access ports that are in use on both switches.
S1(config)# interface range fa0/5 - 6
S1(config-if)# spanning-tree portfast
S2(config)# interface fa0/18
S2(config-if)# spanning-tree portfast
2. Enable BPDU guard on S1 and S2 VLAN 10 access ports connected to PC-A and PC-B.
S1(config)# interface fa0/6
S1(config-if)# spanning-tree bpduguard enable
S2(config)# interface fa0/18
S2(config-if)# spanning-tree bpduguard enable
3. Verify that BPDU guard and PortFast are enabled on the appropriate ports.

S1# show spanning-tree interface fa0/6 detail
Port 8 (FastEthernet0/6) of VLAN0010 is designated forwarding
Port path cost: 19,210,000, Port priority: 128, Port role: designated forwarding
Output only: 1,000,000,000, Port state: 1
Number of times the port has been in the state: 1
The port is in the spanning-tree mode.
Link type is point-to-point by default

Verify that BPDU guard and PortFast are enabled on the appropriate ports.

The screenshot shows two terminal windows and a web browser. The left terminal window (S1) displays the configuration for interface fa0/6, showing that BPDU guard is enabled. The right terminal window (S2) displays the configuration for interface fa0/18, showing that BPDU guard is enabled. The web browser shows a document with a 'Conclusion section' and a 'Step 6: Implement PortFast and BPDU guard' section. The document also includes a table of bindings and a 'Step 7: Verify end-to-end connectivity' section.

```
S1#
S1#show spanningfa0/6 int
* Invalid input detected at '^' marker.
S1#show spanning int fa0/6 int
* Invalid input detected at '^' marker.
S1#show spanning int fa0/6
-----
Vlan      Role  Sts Cost      Prio.Nbr Type
-----
VLAN0001  Desg  FWD  19        128.6   P2p
VLAN0010  Desg  FWD  19        128.6   P2p
VLAN0333  19    128.6   P2p
VLAN0999  19    128.6   P2p
S1#
```

```
S2 (config-if)#
S2 (config-if)#
S2 (config-if)#spanning-tree bpduguard enable
S2 (config-if)#
S2 (config-if)#
S2 (config-if)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console
S2#
S2#show spanning int fa0/18
-----
Vlan      Role  Sts Cost      Prio.Nbr Type
-----
VLAN0001  Desg  FWD  19        128.18  P2p
VLAN0010  Desg  FWD  19        128.18  P2p
VLAN0333  19    128.18  P2p
VLAN0999  19    128.18  P2p
S2#
```

3. **Conclusion section**, written in your own words, that captures a reflective summary of what you have learned and your experience.

Step 6: Implement PortFast and BPDU guard.

1. Configure PortFast on all the access ports that are in use on both switches.
2. Enable BPDU guard on S1 and S2 VLAN 10 access ports connected to PC-A and PC-B.
3. Verify that BPDU guard and PortFast are enabled on the appropriate ports.

S1# show spanning-tree interface fa0/6 detail
Port 8 (FastEthernet6/6) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.6
Designated root has priority 32768, address 0000.A3CD.4183
Designated bridge has priority 32768, address 0000.A3CD.4183
Designated port id is 128.6, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default

S1#

S2#

S2# show spanning-tree interface fa0/18 detail
Port 18 (FastEthernet0/18) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.18
Designated root has priority 32768, address 0000.A3CD.4183
Designated bridge has priority 32768, address 0000.A3CD.4183
Designated port id is 128.18, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default

S2#

3. **Conclusion section**, written in your own words, that captures a reflective summary of what you have learned and your experience.

Step 7: Verify end-to-end connectivity.

Verify PING connectivity between all devices in the IP Addressing Table. If the pings fail, you may need to disable the firewall on the PC hosts.

Close configuration window

Questions to answer

1. In reference to Port Security on S1, what is the maximum value for the maximum age in minutes when sticky learning was configured?

detail

The screenshot shows two terminal windows and a web browser. The left terminal window (S1) displays the configuration for interface fa0/6, showing that BPDU guard is enabled. The right terminal window (S2) displays the configuration for interface fa0/18, showing that BPDU guard is enabled. The web browser shows a document with a 'Conclusion section' and a 'Step 6: Implement PortFast and BPDU guard' section. The document also includes a table of bindings and a 'Step 7: Verify end-to-end connectivity' section.

```
S1#
S1#show spanning int fa0/6 detail
-----
Port 6 (FastEthernet0/6) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.6
Designated root has priority 32768, address 0000.A3CD.4183
Designated bridge has priority 32768, address 0000.A3CD.4183
Designated port id is 128.6, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
S1#
```

```
S2 (config-if)#
S2 (config-if)#spanning-tree bpduguard enable
S2 (config-if)#
S2 (config-if)#spanning-tree bpduguard enable
S2 (config-if)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console
S2#
S2#show spanning int fa0/18 detail
-----
Port 18 (FastEthernet0/18) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.18
Designated root has priority 32768, address 0000.A3CD.4183
Designated bridge has priority 32768, address 0000.A3CD.4183
Designated port id is 128.18, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
S2#
```

3. **Conclusion section**, written in your own words, that captures a reflective summary of what you have learned and your experience.

Step 6: Implement PortFast and BPDU guard.

1. Configure PortFast on all the access ports that are in use on both switches.
2. Enable BPDU guard on S1 and S2 VLAN 10 access ports connected to PC-A and PC-B.
3. Verify that BPDU guard and PortFast are enabled on the appropriate ports.

S1# show spanning-tree interface fa0/6 detail
Port 8 (FastEthernet6/6) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.6
Designated root has priority 32768, address 0000.A3CD.4183
Designated bridge has priority 32768, address 0000.A3CD.4183
Designated port id is 128.6, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
Bpdu guard is enabled
Bpdu: sent 128, received 0

S1#

S2#

S2# show spanning-tree interface fa0/18 detail
Port 18 (FastEthernet0/18) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.18
Designated root has priority 32768, address 0000.A3CD.4183
Designated bridge has priority 32768, address 0000.A3CD.4183
Designated port id is 128.18, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
Bpdu guard is enabled
Bpdu: sent 128, received 0

S2#

3. **Conclusion section**, written in your own words, that captures a reflective summary of what you have learned and your experience.

Step 7: Verify end-to-end connectivity.

Verify PING connectivity between all devices in the IP Addressing Table. If the pings fail, you may need to disable the firewall on the PC hosts.

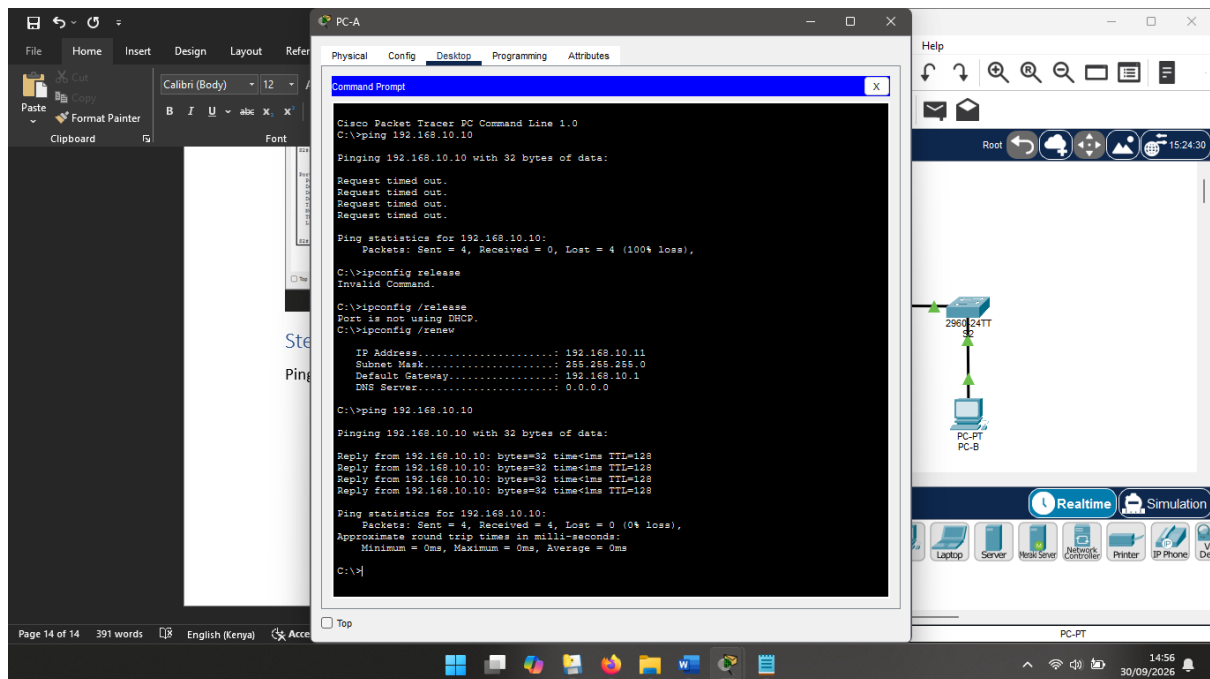
Close configuration window

Questions to answer

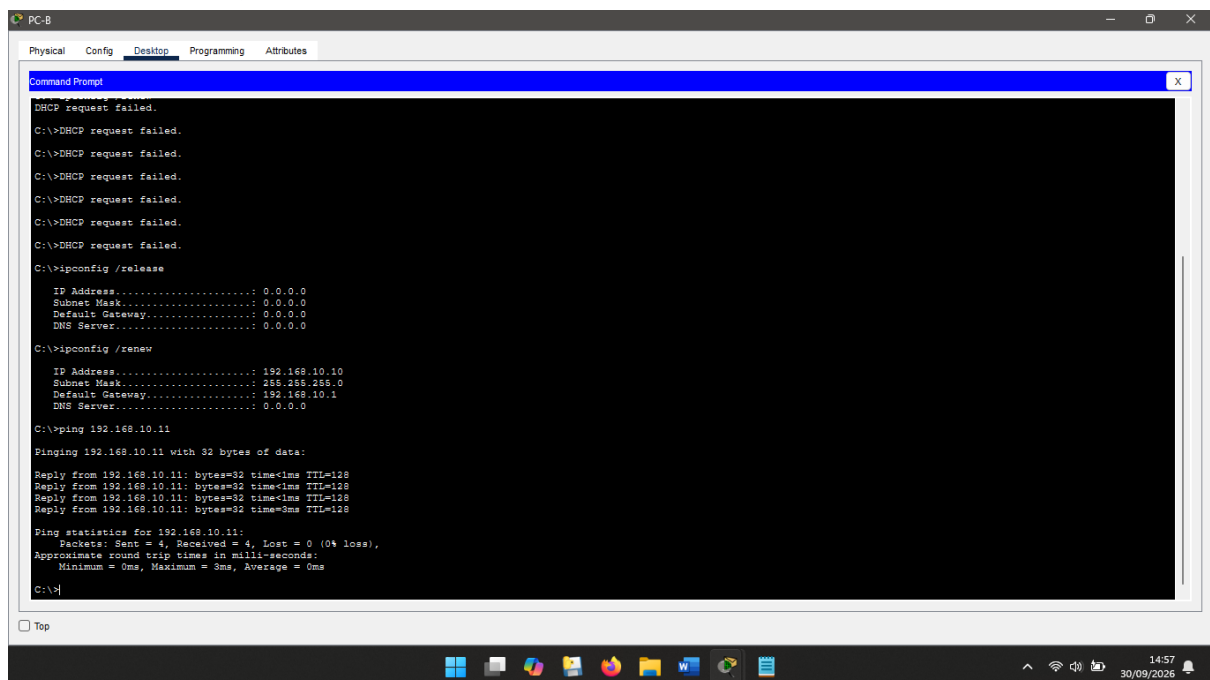
1. In reference to Port Security on S1, what is the maximum value for the maximum age in minutes when sticky learning was configured?

Step 7: Verify end-to-end connectivity.

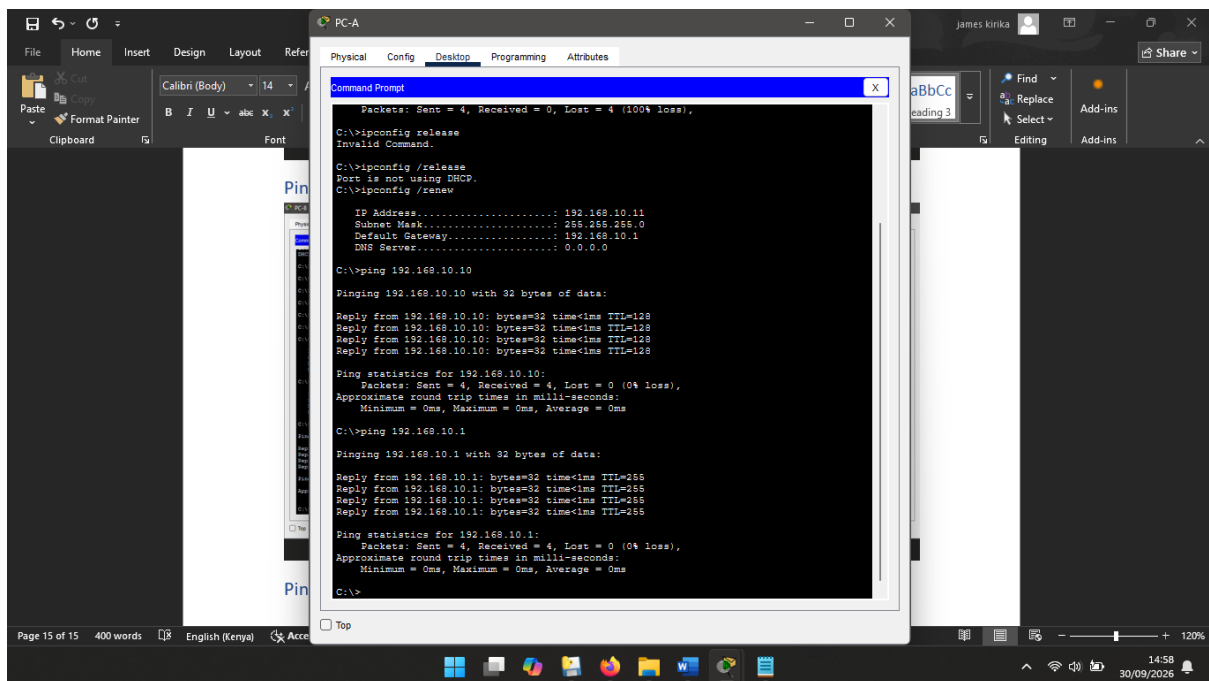
Ping pcb B from pc A



Ping pc A from pc B



Ping default gateway



The screenshot shows a Windows 10 desktop environment. A Command Prompt window is open, displaying the following commands and output:

```
C:\>ipconfig /release
Invalid Command.

C:\>ipconfig /release
Port is not using DHCP.
C:\>ipconfig /renew

IP Address. . . . . : 192.168.10.11
Subnet Mask. . . . . : 255.255.255.0
Default Gateway. . . . . : 192.168.10.1
DNS Server. . . . . : 0.0.0.0

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

The background shows a Microsoft Word document titled 'Pin' and a Windows taskbar at the bottom with the system clock showing 14:58 on 30/09/2026.

Conclusion

This lab showed how a basic switched network can be segmented and hardened at Layer 2 using built-in switch features. VLAN 10 was created for management and user traffic, with 802.1Q trunking between S1 and S2 using VLAN 333 as an unused native VLAN, and DTP negotiation disabled to prevent trunk manipulation. Unused ports were assigned to VLAN 999 and shut down to block unauthorized access. Port security limited the MAC addresses allowed on the PC-facing ports, using different violation modes and aging settings on S1 and S2. DHCP snooping with rate limiting protected against rogue DHCP servers and DHCP flooding, while PortFast and BPDU guard sped up host connectivity and shielded the spanning-tree topology from rogue switches. End-to-end connectivity tests confirmed that these controls secured the network without breaking legitimate traffic.